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Theoretical study on the effect of oxidation states of phosphorus flame retardants on their mode of action

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ABSTRACT

Polymers are ubiquitous and find applications in various facets of daily life and industrial production. However, the fire incidents induced by polymers jeopardize the safety of human life and property. Improving the fire resistance of polymeric materials through halogen-free flame retardants has become a prominent direction guided by the regulatory framework. In this paper, the decomposition pathways and mechanisms of five phosphorus-containing flame retardants, PPh3, DPPO, PDPP, TPPO and TPP, were investigated. The reaction pathways primarily encompassed the generation of PO, PO_2, PO_3, PO_4, phenyl, and phenoxy radicals. Through a comprehensive analysis of the dissociation energies of P-C, P-O, and C—O bonds under diverse chemical environments, phenoxy radicals was the principal decomposition products, with the cleavage of P-O bonds being the predominant choice. Increasing phosphorus oxidation states favors the accumulation of a carbon layer, tending to retard flame propagation through condensed-phase mechanisms. Phosphorus oxidation states primarily facilitated secondary dissociation reactions. The presence of oxygen favored the cleavage of P-C bonds, offering diverse flame retardant mechanisms that concurrently mitigated flame spread. This work provided an in-depth exploration of the flame retardant mechanisms of phosphorus-containing agents and summarized the impact of phosphorus oxidation states on flame retardants.

1. Introduction

Polymer materials, characterized by their lightweight properties, exceptional stability, ease of processing, and resistance to corrosion, exhibit diverse applications in both everyday life and industrial settings [1,2]. These applications are found in various sectors, including medicine, automotive, aerospace, electronics, and transportation. However, the intrinsic chemical structure and organic components of polymer materials contribute to their highly flammable [3,4]. The considerable loss of life and property caused by fires globally each year further constrains the practical applications of these materials [5]. Enhancing the fire resistance of polymer materials has consequently become the primary focus of researchers [6].

The introduction of flame retardants has provided a new avenue for addressing these issues [7,8]. By chemically incorporating appropriate flame retardant into polymers, the fire spread can be thwarted through processes such as heat absorption, oxygen isolation, interruption of chemical chain reactions, and the generation of non-combustible gases [9]. Traditional flame retardants mainly consist of halogenated flame retardants (such as brominated and chlorinated flame retardants), which possess highly effective abilities to suppress flame propagation. During the degradation process, however, they generate highly toxic substances like hydrogen halides harmful to organisms and the environment [10,11]. Furthermore, due to regulatory requirements, the development of halogenated flame retardants is restricted in production, which has stimulated scientists to explore non-halogenated flame retardants. Phosphorus-containing flame retardants, as a highlight in the non-halogenated flame retardant domain, present unique advantages such as high efficiency and multiple flame retardant mechanisms while meeting safety and green environmental requirements [12,13]. According to the HIS Consulting report in 2020, phosphorus-containing flame retardants account for 18 % of the world's flame retardant consumption [14]. To design and development more effective phosphorus-containing flame retardants has practical value [15].

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Moreover, gaining insight into the combustion process of polymers is beneficial for further exploring the combustion mechanisms of phosphorus-containing flame retardants. The combustion process can be simply distilled into the tetrahedron of fire: oxygen, fuel, temperature and chain reaction. The reactions of the combustion process are predominantly the following two processes:

$$HO\cdot + CO \rightarrow CO_2 + H\cdot \quad (1)$$

$$H\cdot + O_2 \rightarrow HO\cdot + O\cdot \quad (2)$$

Designing flame retardants based on the above four elements can effectively reduce the combustibility of materials. Previous studies have shown that phosphorus-containing flame retardants mainly function in the condensed phase and gas phase of polymer combustion. The carbon formation containing P-O-P and P-O-C compounds produced by the aluminum-supported branched oligo(phenylphosphate) designed by Yuan et al. during combustion effectively insulated the entry of oxygen and prevented the temperature from rising, by working on the condensed phase of the fire tetrahedron to destroy it [16]. In the gas phase, Shmakov and his colleagues [17] investigated the impact of hexabromocyclododecane (HBCD) and triphenyl phosphate oxide (TPPO) on $CH_4/O_2/N_2$ flames. Experimental results indicated that both TPPO and HBCD suppressed the flame by reducing the concentration of H· and OH· free radicals. Pham et al. [18]. studied the influence of triphenyl phosphate (TPP) on the thermal and flame-retardant properties of polypropylene/wood flour (PP-WF). Thermal analysis (TG) and Fourier-transform infrared spectroscopy (FTIR) characterization results demonstrated that TPP released phosphorus-containing radicals during combustion to capture H· and OH· free radicals, interrupting the chain reaction of combustion. Korobeinichev and his colleagues [19] explored the flame-retardant mechanism of TPP in the combustion of polymethyl methacrylate through experiments and quantum chemical calculations. It confirmed that TPP produced small phosphorus oxide molecules, such as PO and PO_2. These studies didn't provide decomposition pathways and detailed reaction mechanisms of phosphorus-containing flame retardants when combustion. Interestingly, the majority of the literature focused on the enhancement of flame retardant properties by incorporating phosphorus-containing flame retardants into polymers, such as ammonium polyphosphate (APP) [20], DOPO-ABZ [21], TPP [4], 2-carboxyethyl(phenyl)phosphinic acid (CEPPA) [22]. Only a few researchers have investigated the influence of different phosphorus oxidation states on the flame retardant mechanisms of polymers. Based on this, we selected five flame retardants with different oxidation states, including triphenylphosphine (PPh3), TPPO, phenyl diphenylphosphinate (PDPP), diphenyl phosphate (DPPO), and TPP, to explore the potential reaction mechanisms and detailed chemical kinetics during the combustion process. PPh3 is a relatively strong nucleophilic reagent, and the presence of phosphorus gives the molecule a certain degree of stability. It is often used as an important intermediate in a variety of chemical reactions. Phosphorus contents of up to 11.8 % can increase the limiting oxygen index of polymer materials. When PPh3 is used as a flame retardant with epoxy resins, it has a high rate of carbon formation, is not easily decomposed, and exhibits a condensed-phase mechanism to quench the flame. Exploring the pyrolysis mechanism of PPh3 flame retardants can help to further understand the flame retardant process of PPh3 and polymer composites based on PPh3 derivatives.

The aim of this work was to comprehensively explore the flame retardant mechanism of the phosphorus oxidation states on polymers. Employing ab initio quantum chemistry methods, we endeavored to construct precise computational models and utilized high-precision calculation techniques to unveil the intricate decomposition pathways and reaction mechanisms of phosphorus flame retardants. The energy and potential barriers of the 27 reaction pathways, constituting the dissociation of P-C, P-O, and C—O bonds under different chemical environments, were systematically analyzed. The increase of phosphorus oxidation states facilitated the secondary reaction of P-C bond cleavage. The final decomposition steps became challenging with the phosphorus oxidation states rose, favoring the implementation of condensed-phase mechanisms for flame extinguishment. This research further enhanced the accuracy of the dynamic models for phosphorus oxidation states during the polymer combustion process, which will provide deeper insights into flame retardant design and lay the foundation for designing more efficient flame retardant materials.

2. Theoretical calculation

Fig. 1 illustrates the optimized molecular structures for PPh3, TPPO, PDPP, DPPO, and TPP, designated as RC1, RC2, RC3, RC4, and RC5, respectively, in the reactions. The decomposition pathways of each reaction product are contingent upon the cleavage of P-C, P-O, and C—O bonds, resulting in the formation of C_6H_5 and C_6H_5O radicals. The detailed mechanistic pathway diagram is shown in Fig. S1 in the supplementary material. Notably, RC1 and RC2 exclusively involve the rupture of P-C bonds, whereas RC3, RC4, and RC5, characterized by elevated oxidation states of phosphorus, facilitate generation of C_6H_5O radicals through the cleavage of P-O bonds. Drawing on precedent studies, it is established that phosphorus-containing flame retardants dissociate at elevated temperatures, yielding POx ($x = 1, 2, 3, 4$) radicals to capture H· and OH· radicals. A comprehensive exploration of the decomposition mechanisms associated with phosphorus in varying oxidation states is helpful to understand the functionality of phosphorus-containing flame retardants.

All structural optimizations in this study were conducted employing the 6–31G(d,p) basis set and the M06–2X method. M06–2X, a first-principles computational approach tailored for thermodynamic and kinetic assessments, provided structures, potential energy surfaces (PES), and frequencies of relative precision within permissible bounds. Concomitantly, frequency calculations were performed at an equivalent theoretical level, ensuring the absence of imaginary frequencies for reactants, intermediates, and products, thus assuring accurate determination of zero-point vibrational energies (ZPE) for correction. All scans of single bonds for the reactants and intermediates were executed using the M06–2X method with a dihedral angle increment of 10° until the completion of the 360-degree rotational potential. The resulting data were conventionally employed for hindered rotor corrections, pivotal for rate constant determinations. The structural conformers with the lowest energy were selected for each reactant and intermediate.

In accordance with reported literature, refinement of the G3 chemical model was deemed beneficial for enhancing the accuracy of molecular energies for phosphorus-containing species, thereby augmenting the theoretical foundation of this investigation [23]. Single-point energy computations for each species were conducted utilizing the G3(MP2, CC) method, supplemented by ZPE corrections [24]. The corrected second-order Møller–Plesset perturbation theory (MP2) and G3Large were integrated into the coupled cluster CCSD(T)/6–31G(d,p) framework. The anticipated computational results had an error within 2 kJ/mol compared to literature for analogous species [25]. The reaction rate constants depended on the reaction barriers [26]. Dissociation curves for primary reaction pathways involving P-C, P-O, and C—O were computed, and the frequencies and hindered rotor corrections for each structure were calculated based on point sampling along the dissociation curves. The detailed structural parameters obtained from the calculations of reactants and intermediates were undertaken in the text to determine the rate constants of the reaction channels. All calculations in this work were performed using the Gaussian16 software [27].

3. Results and discussion

Fig. 2 demonstrates all the reaction channels presented in the combustion process of RC1, RC2, RC3, RC4 and RC5. Due to the structural homogeneity of these reactants, the same intermediates are produced during the fracture of P-C, P-O, and C—O bonds. No more separate

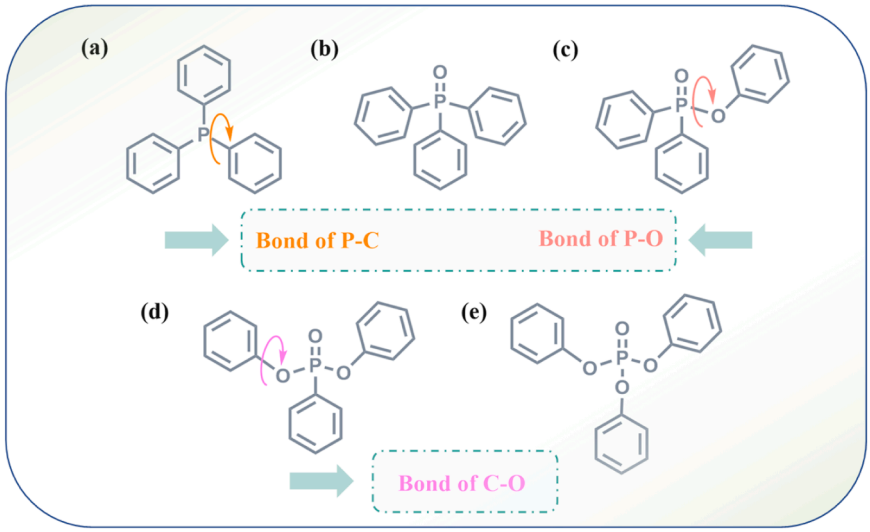

Fig. 1. Schematic structures of the five researched materials, (a) PPh3, (b) TPPO, (c) PDPP, (d) DPPO, and (e) TPP, with the three types of bond breaks including in the decomposition pathway, P-C, P-O, and C—O, respectively.

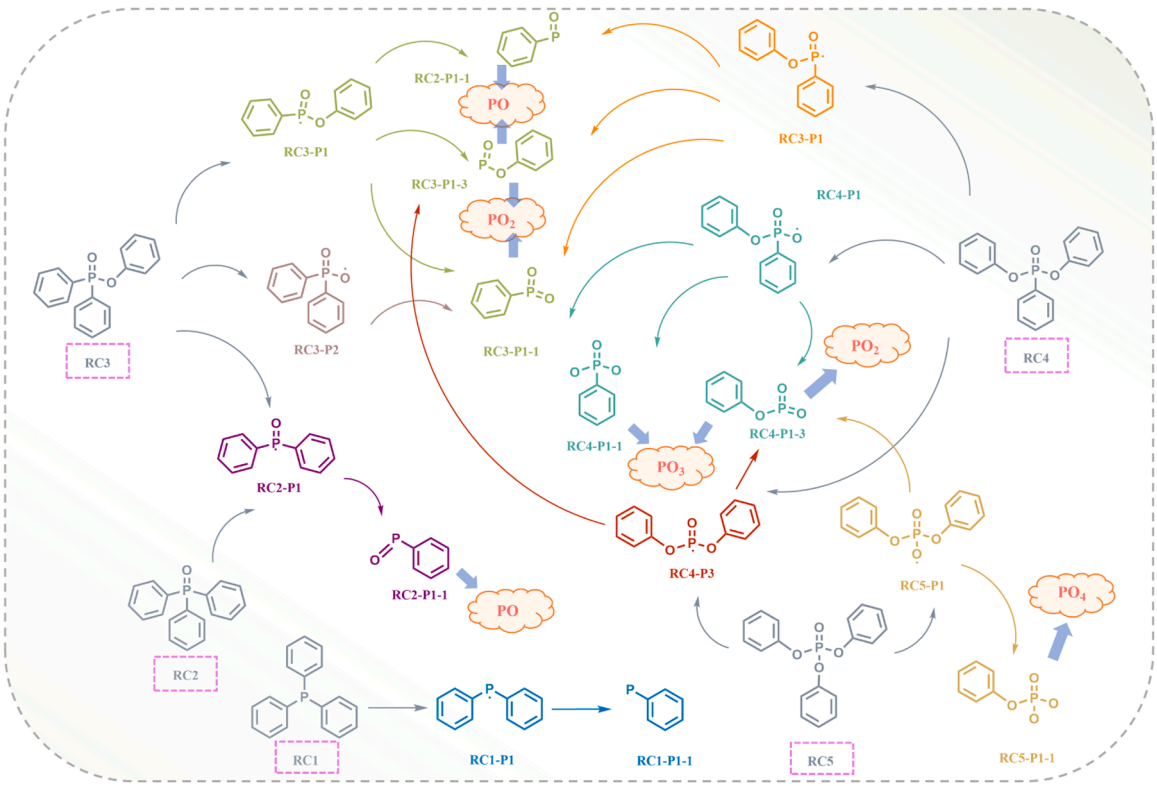

Fig. 2. Diagram of all decomposition pathways for the five reactants, with multiple colors representing different reaction pathways.

designations for the same intermediate. Structures in gray represent primary reactants, different colors represent different reaction paths. Obviously, RC1 and RC2 exhibited exclusive decomposition channels, while RC3 and RC4 decompose through three discrete pathways contingent upon the rupture of P-C, P-O, and C—O bonds. With the maximal number of oxygen atoms and a highly symmetric structure, RC5 follows exclusively two reaction pathways governed by the cleavage of P-O and O—C bonds. This is also true for the reaction pathway of the intermediate, which end with the decomposition into PO, PO_2, PO_3 and PO_4 (including phenyl and phenoxy radicals). As can be clearly seen in Fig. 2, RC2-P1–1, RC3-P1–1, RC3-P1–3, RC4-P1–1, RC4-P1–3, and RC5-P1–1 are the ultimate intermediates. Each reaction pathway is derived from the successive dissociation order of benzyloxy radicals and benzyl radicals within distinct chemical environments. The reactions involving phosphorus radicals, generated by flame retardants in diverse oxidation states, capturing H· and OH· radicals are as follows:

$$PO\cdot + H\cdot \rightarrow HPO \tag{3}$$

$$PO\cdot + OH\cdot \rightarrow HPO_2 \tag{4}$$

$$HPO + H\cdot \rightarrow PO\cdot + H_2 \tag{5}$$

$$PO\cdot + OH\cdot + H_2 \rightarrow HPO_2\cdot + H_2O \tag{6}$$

$$HPO_2\cdot + H\cdot \rightarrow PO\cdot + H_2 \quad (7)$$

$$HPO_2\cdot + OH\cdot \rightarrow PO_2\cdot + H_2O \quad (8)$$

Phosphorus radicals are involved in complex reactions with combustion-responsive radicals, leading to the continuous consumption of H· and OH· radicals. Simultaneously, these reactions resulted in the formation of phosphoric acid, PO·, and PO_2· radicals, thereby extinguishing the flame.

Phosphates usually work in condensed phase mechanisms:

$$2H_3PO_4 + 5C \rightarrow 2P + 5CO + 3H_2O \quad (9)$$

$$4P + 5O_2 \rightarrow 2P_2O_5 \quad (10)$$

$$P_2O_5 + 5C \rightarrow 2P + 5CO \quad (11)$$

3.1. Decomposition pathways of PPh3 and TPPO flame retardant

As depicted in Fig. 3, both PPh3 (RC1) and TPPO (RC2) monomers exhibit a singular reaction pathway. RC1 is cleaved through the P-C bond to eventually form RC1-P1 and phenyl radicals, which further evolve into RC1-P1–1 through benzene ring detachment. In TPPO, the 2*C_6H_5 radicals is similarly ensued through P-C bond breaking. Phosphorus flame retardants containing a single oxygen will be further cleaved to produce PO radicals. The primary dissociation energies of PPh3 are slightly lower than TPPO, measuring 91.7 kcal/mol and 102.4 kcal/mol, respectively. Although the primary dissociation energy of TPPO is higher than PPh3, the presence of oxygen during the detachment of the second C_6H_5 markedly favors the further dissociation of TPPO. A substantial quantity of PO radicals is generated and volatilized into the gas phase. Subsequently, upon reacting with OH· and H· radicals in the flame, phosphorus-containing compounds and phosphorus oxyacids are generated, effectively inhibiting the spread of fire and manifesting a gas-phase flame retardant effect.

3.2. Decomposition pathways of PDPP flame retardant

As illustrated in Fig. 4, the PDPP monomer undergo P-C bond cleavage, resulting in the dissociation into the RC3-P1 intermediate, followed by the shearing of the C—O bond to form RC3-P2. The intermediate is structurally identical to RC2-P1 through the cleavage of the P-O bond. Significantly, the P-O bond exhibits the lowest susceptibility to cleavage (90.0 kcal/mol), followed by the P-C bond (105.6 kcal/mol) and the C—O bond (116.2 kcal/mol), which implies that the pathway leading to the generation of RC3-P2 and benzyl radicals through the C—O bond is not competitive. Within this pathway, the intermediate RC3-P1 undergo three distinct reaction pathways through the shearing

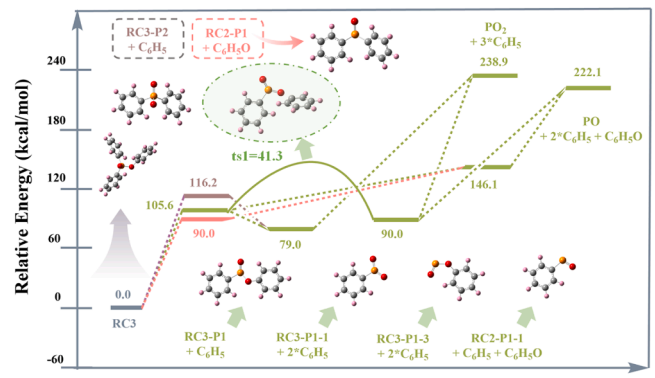

Fig. 4. Reaction channel and energy distribution diagram of DPPO flame retardant.

of three types of bonds. By comparing the four possible reaction pathways of RC3-P1, it can be observed that the cleavage of the P-O bond remains the most favorable reaction pathway, with a calculated reaction energy of 76.0 kcal/mol. In summary, the pathway solely involving the cleavage of the C—O bond is hindered due to the high bond energy. The reaction pathways of RC3-P1 intermediate are more complex, and a comparative analysis of dissociation energies for each reaction pathway revealed that the P-O bond governed the flame-retardant properties. In the mechanistic study of the PDPP flame retardant, the dominance of the reaction involving PO + benzyloxy radicals + benzyl radicals is established. The reaction pathway RC3 → RC2-P1 + C_6H_5O → RC2-P1–1 + C_6H_5 + C_6H_5O → PO + C_6H_5 + C_6H_5O is the principal mechanism. In addition, it is noteworthy that the dissociation energies of the tertiary reactions are consistently lower than those required for the generation of RC3-P1 through primary reactions. This observation implies that the incidence of primary reactions still promotes the generation of phosphorus oxygen radicals. However, only minuscule quantities of intermediates may not proceed to sequential decomposition steps, culminating in the formation of a carbonaceous layer during combustion.

3.3. Decomposition pathways of DPPO flame retardant

Referring to Fig. 5, during the primary reaction, RC4 undergo three distinct reaction pathways via bond cleavage. The successive generation of phenyl necessitated the rupture of C—O and P-C bonds. The energy required for the primary reaction is the highest, reaching 111.5 and 111.4 kcal/mol. In contrast, the secondary reaction through the decomposition of the P-C bond to form RC4-P1–3 is more favorable than RC3-P1–1. However, the reaction energy for RC4-P1–3, leading to PO_3 formation, surpasses that of RC3-P1–1, which favors the formation of

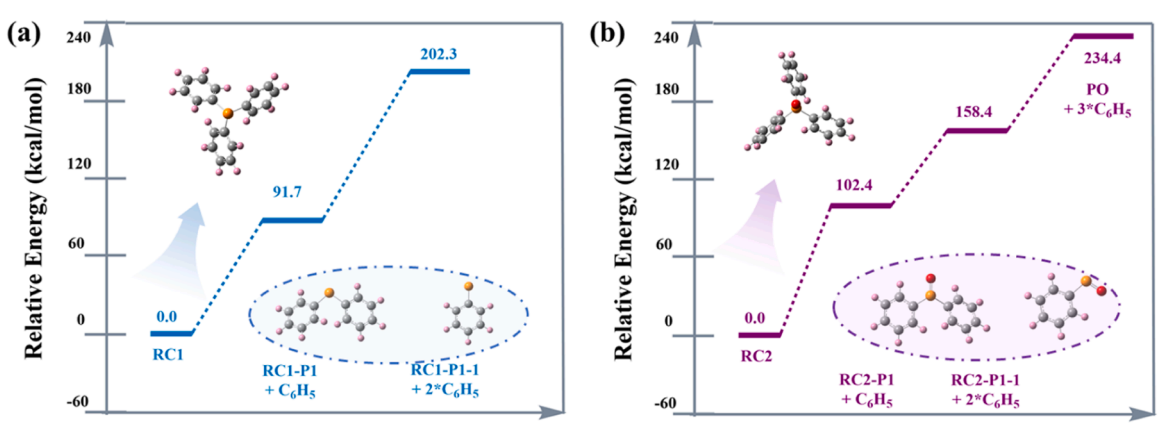

Fig. 3. Reaction channel and energy distribution diagram of PPh3 and TPPO flame retardant.

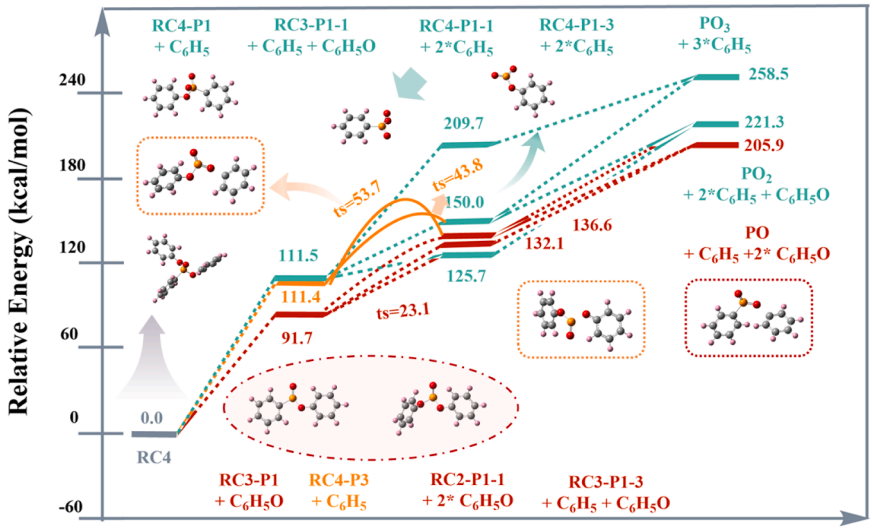

Fig. 5. Reaction channel and energy distribution diagram of DPPO flame retardant.

PO_2 with a reaction energy of 71.3 kcal/mol. Notably, the dissociation energy for phenoxide radical generation in the secondary reaction is remarkably low at 14.2 kcal/mol. Despite the higher dissociation energy for PO_2 compare to the preceding intermediates, comprehensive analysis deems its competitiveness.

Another pathway involving the generation of phenyl radicals in the primary reaction is the formation of RC4-P3. The intermediate formed after the rupture of the C—O bond are consistent with the pathway of RC3-P1–1. However, cleavage of the P-O bond leads a lower dissociation energy (25.2 kcal/mol), rendering the decomposition to produce PO + phenyl radicals more favorable. Surprisingly, the pathway involving the first-time rupture of the P-O bond in the primary reaction exhibits a more substantial reaction energy, significantly lower than the other two intermediates. In the subsequent second and third order reaction, the P-O bond or P-C bond has a lower reaction energies. In comparison, the pathway favoring dissociation through the initial cleavage of the P-C bond, followed by the rupture of the P-O bond leading to the formation of PO + phenyl radicals and phenoxide radicals, proved more advantageous. Hence, throughout the combustion process, the flame-retardant mechanism of DPPO is RC4 → RC3-P1 + C_6H_5O → RC3-P1–3 + C_6H_5 + C_6H_5O → PO + C_6H_5 + 2*C_6H_5O. DPPO, endows with three oxygen atoms, exhibits diverse phosphorus oxidation states, imparting a complexity to the flame-retardant mechanism. The concurrent engagement of multiple flame-retardant mechanisms not only eradicate active radicals but also obstruct the temperature transfer from the combustion zone to the condensed phase, slowing down the flame propagation. The dissociation energy involved in the ultimate step generally surpasses that of secondary reactions. At lower combustion temperatures, the phosphorus-containing intermediates do not further dissociate, but form a dense carbonaceous residue that enriches in the polymer material surface. This dense layer served as an insulating barrier, effectively imparting flame-retardant by preventing oxygen ingress. Conversely, at heightened combustion temperatures, the resultant PO_3 reacts with active radicals in the flame gas phase to deliver pyrophosphate. Pyrophosphate possesses robust dehydration properties, and the formed thin film is conducive to the generation of carbon layer. Therefore, the decomposition pathway includes the liberation of phosphorus-containing volatiles, while a fraction of phosphorus persists in the condensed phase due to branching reactions requiring high dissociation energy.

3.4. Decomposition pathways of TPP flame retardant

In Fig. 6, the decomposition pathways that potentially occur during

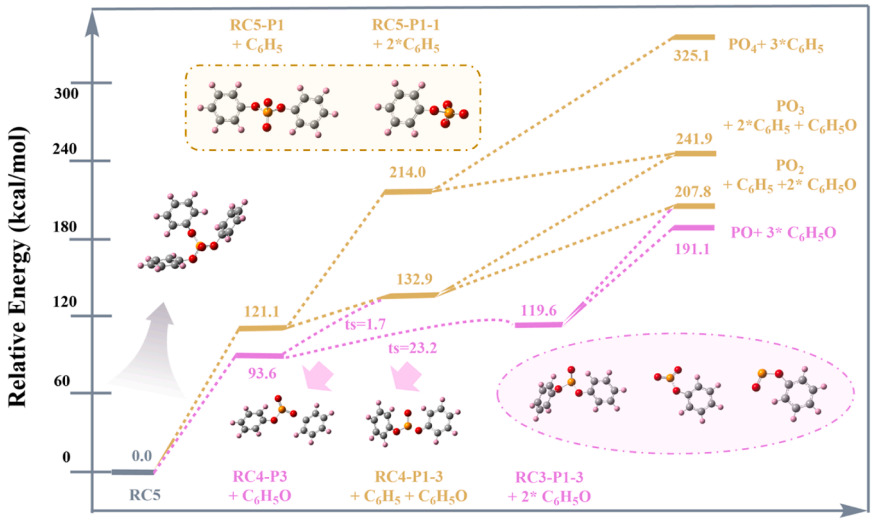

Fig. 6. Reaction channel and energy distribution diagram of TPPO flame retardant.

combustion of the TPP (RC5) flame retardant were illustrated. Due to the structural symmetry of RC5, the chemical environment of the three phenoxide groups is identical, leading to the generation of quenching radicals PO, PO_2, PO_3, and PO_4 during combustion and releasing into the gas phase. The primary dissociation reactions involve the breaking of P-O and C—O bonds. Sequential C—O bond cleavage exhibits the highest reaction energy, with dissociation energies of 121.1, 92.9, and 111.1 kcal/mol in each step. The weakest bond is the P-O bond and when cleaved, RC5-P1–1 decomposed into PO_3, phenyl radicals, and phenoxide radicals. The RC5-P1 intermediate can also generate RC4-P1–3 through the cleavage of P-O bonds, with notably lower energy than RC5-P1–1, indicating greater selectivity in the reaction. The tertiary reactions that occur is similar to that of DPPO, with the primary decomposition of RC5 yielding RC4-P3 as the minimum dissociation energy product. RC4-P3 formation favors phenoxide radicals (26.0 kcal/mol) over phenyl radicals (39.3 kcal/mol). Comparative analysis of all reaction pathways, the reaction path of phenoxy radical elimination is the main path, and the decomposition product is $PO_2 + C_6H_5 + 2C_6H_5O$. The reaction is described as RC5 → RC4-P3 + C_6H_5O → RC4-P1–3 + C_6H_5O + C_6H_5 → PO_2+ C_6H_5 + 2*C_6H_5O. Similarly, the dissociation energy in the final step exceeded that of secondary reactions. Under lower fire intensities, TPP decompose into phosphorus-containing intermediates covering the polymer surface, manifesting as a condensed-phase mechanism. When the temperature is high enough, these intermediates further form small phosphorus fragments and dehydrating phosphoric and pyrophosphoric acid, capturing combustion radicals while forming a dense carbon layer that effectively suppresses the flame.

3.5. Analysis of bond energies of PPh3, TPPO, PDPP, DPPO and TPP flame retardants with respect to P-C, P-O and C-O

To further analyze the effect of different phosphorus oxidation states on the flame retardant mechanism of polymers, we summarized the changes of P-C, P-O and C—O bond energies of PPh3, TPPO, PDPP, DPPO and TPP. As shown in Fig. 7(a), the increase of phosphorus oxidation states in the primary reaction is beneficial for C—O bond breaking of DPPO, while the P-C and P-O bond energies are in an upward trend, with the most pronounced increase in the P-C bond. In Fig. 7(b),

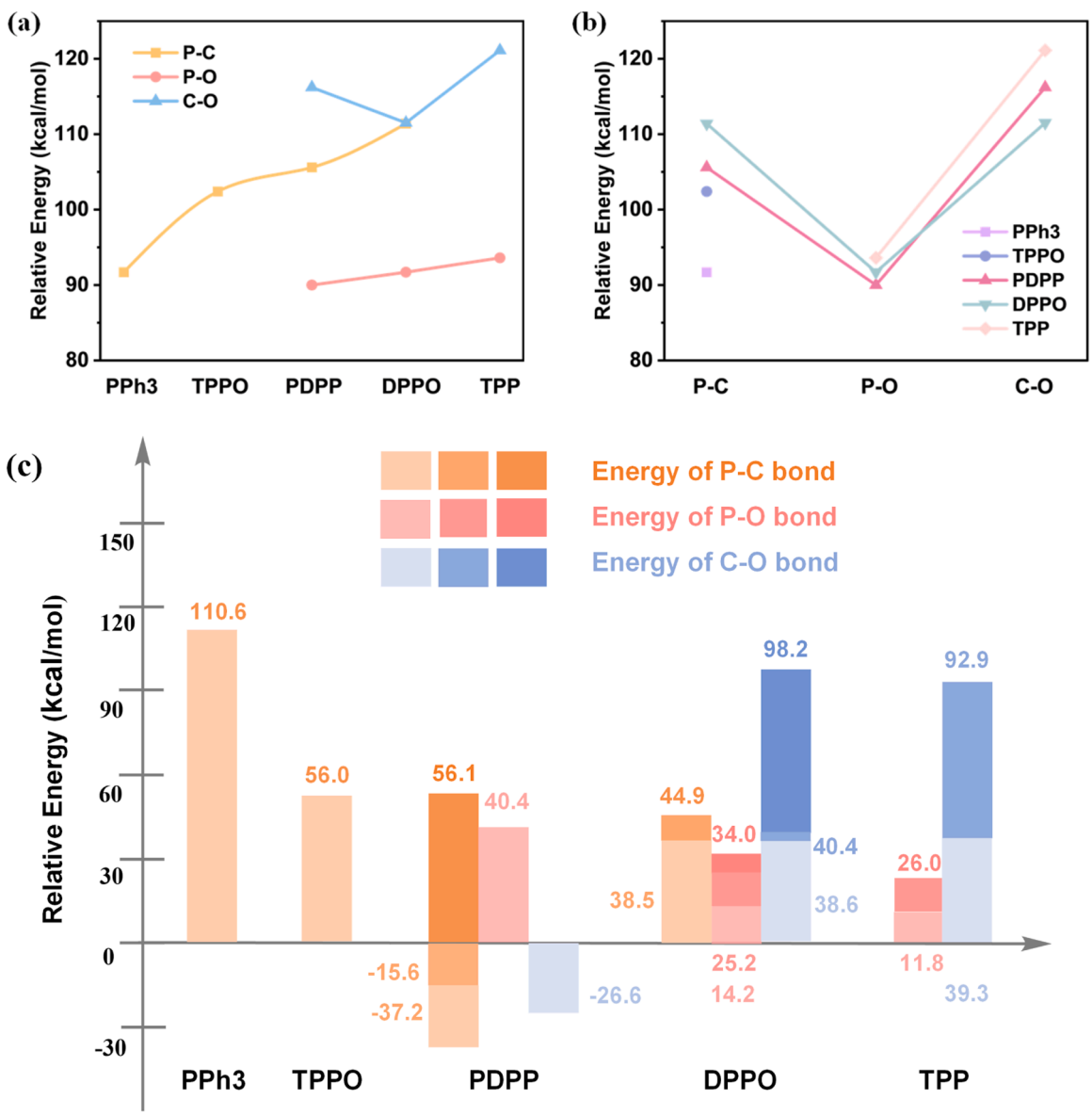

Fig. 7. (a) The distribution of P-C, P-O and C—O bond energies in the primary reaction of PPh3, TPPO, PDPP, DPPO and TPP; (b) Energy distribution of P-C, P-O and C—O bonds for PPh3, TPPO, PDPP, DPPO and TPP, respectively. (c) Energy distribution of P-C, P-O and C—O bonds in the secondary reaction in PPh3, TPPO, PDPP, DPPO and TPP.

the P-O bond is located in the deepest position, which suggests that phenoxy displays a higher selectivity in primary reactions. The cleavage of the secondary reaction is more complex (see shown in Fig. 7(c)). For the P-C bond, an increase in the phosphorus oxidation states favors the occurrence of secondary cleavage. The P-O bond is preferred for the pyrolysis reactions of PDPP, DPPO, and TPP, where high dissociation energies are virtually non-existent. In contrast, C—O bond segregation is extremely difficult to occur for TPP and DPPO flame retardants. In the reactions generating phosphorus-oxygen fragment reveals that most of the tertiary reactions had much higher energies than the secondary reactions (see Fig. S2 in the supplementary material). And with the phosphorus oxidation states increase, the hindering effect increases and tends to form a carbon layer. During polymer combustion, flame retardants in higher phosphorus oxidation states predominantly reside in intermediates generated by secondary reactions, which form a film covering the polymer surface when heat is absorbed enough. High temperature will cross the dissociation energy of the tertiary reaction to decompose into phosphorus-containing oxide fragments that traps reactive radicals in the gas phase. The rate constants for the three types of bonds are shown in Fig. 8 and the results are more in line with the literature [22]. The details of the specific calculations are in Supplementary Material.

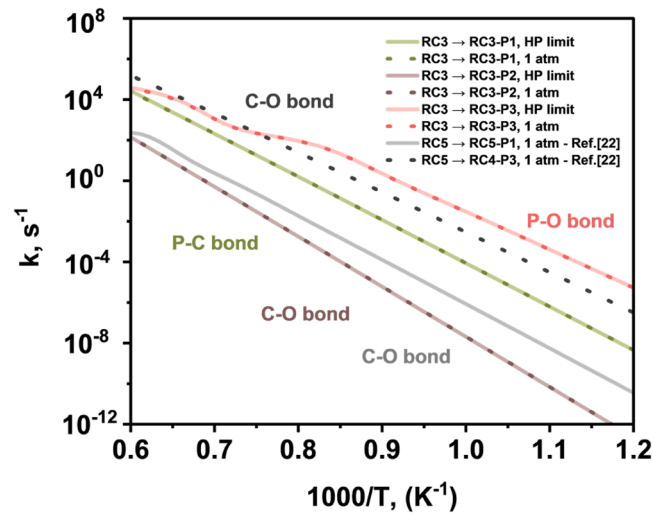

Fig. 8. Rate constants for P-O, P-C and C—O bonds in the 1 atm and the high-pressure limit (HP) compared to the literature for the same type of bond.

4. Conclusions

To scrutinize the effect of phosphorus oxidation states on the flame retardant mechanism of polymers, the mechanisms of five phosphorus-containing flame retardants, PPh3, DPPO, PDPP, TPPO, and TPP, were investigated. The cleavage of P-C, P-O, and C—O bonds constituted 27 reaction paths for the five reactants. The main decomposition pathways and mechanisms of each substance were proposed based on the dissociation energy of each reaction pathway, and the cleavage laws of P-C, P-O and C—O were analyzed. Phosphorus oxidation provided a variety of flame retardant mechanisms, and the following results were obtained.

(1) TPPO inhibited flame mainly by gas-phase mechanism, with only trace amounts of intermediates will form a dense carbon layer, which was manifested as cohesive-phase flame retardancy. And with the increase of phosphorus oxidation states, the condensed phase mechanism acts as a dominant role.
(2) In our exploration of the flame retardant mechanism of PDPP, the principal reaction pathway involved RC3 → RC2-P1 + C_6H_5O → RC2-P1–1 + C_6H_5 + C_6H_5O → PO + C_6H_5 + C_6H_5O. For the DPPO flame retardant, the most favorable flame retardant mechanism was RC4 → RC3-P1 + C_6H_5O → RC3-P1–3 + C_6H_5 + C_6H_5O → PO + C_6H_5 + 2*C_6H_5O. The reaction mechanism of TPP was RC5 → RC4-P3 + C_6H_5O → RC4-P1–3 + C_6H_5O + C_6H_5 → PO_2 + C_6H_5 + 2*C_6H_5O.
(3) Phenoxy was more selective than phenyl in the flame retardant process, and the P-O bond was a more favorable dissociation site. The trend of the C—O bond during the reaction was not obvious, and the higher dissociation energy indicated that it was not competitive.
(4) The increase in phosphorus oxidation states hindered dissociation in primary reactions but greatly promoted the cleavage of P-C bonds, leading to the generation of secondary products.
(5) Under low fire intensities, the thermal decomposition of C—O bonds in phosphorus-containing flame retardants generated PO_4, PO_3, PO_2, and PO, which volatilized into the gas phase, eliminating flame radicals while generating phosphoric acid and metaphosphoric acid, which effectively promoted the carbon layer.
(6) PDPP and TPPO are more easily dissociated into small phosphorus oxide molecules with quenching effect, followed by TPP, DPPO and PPh3. In the actual fire extinguishing process, the high-valence phosphorus oxidation states exhibit a variety of flame retardant mechanisms, and possesses a remarkable flame retardant effect.

CRediT authorship contribution statement

Chunlan Qin: Investigation, Writing – original draft, Visualization. **Jie Chen:** Data curation, Investigation. **Shanshan Ruan:** Formal analysis. **Fuyi Liu:** Funding acquisition. **Lidong Zhang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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References

[1] L. Liu, J. Feng, Y. Xue, V. Chevali, Y. Zhang, Y. Shi, L.C. Tang, P. Song, 2D MXenes for fire retardancy and fire-warning applications: promises and prospects, Adv. Funct. Mater. 33 (9) (2023) 2212124.
[2] S. Liang, P. Hemberger, M. Steglich, P. Simonetti, J. Levalois-Grützmacher, H. Grützmacher, S. Gaan, The underlying chemistry to the formation of PO2 radicals from organophosphorus compounds: a missing puzzle piece in flame chemistry, Chem. - Eur. J. 26 (47) (2020) 10795–10800.
[3] B.C. Roberts, A.R. Jones, O.A. Ezekoye, C.J. Ellison, M.E. Webber, Development of kinetic parameters for polyurethane thermal degradation modeling featuring a bioinspired catecholic flame retardant, Combust. Flame 177 (2017) 184–192.

[4] M. Kilinc, G.O. Cakal, G. Bayram, I. Eroglu, S. Özkar, Flame retardancy and mechanical properties of pet-based composites containing phosphorus and boron-based additives, J. Appl. Polym. Sci. 132 (22) (2015) 42016.
[5] S. Wang, X. Huang, H. Chen, N. Liu, G. Rein, Ignition of low-density expandable polystyrene foam by a hot particle, Combust. Flame 162 (11) (2015) 4112–4118.
[6] M.M. Velencoso, A. Battig, J.C. Markwart, B. Schartel, F.R. Wurm, Molecular firefighting—how modern phosphorus chemistry can help solve the challenge of flame retardancy, Angew. Chem. Int. Ed. 57 (33) (2018) 10450–10467.
[7] S. Liang, P. Hemberger, N.M. Neisius, A. Bodi, H. Grützmacher, J. Levalois-Grützmacher, S. Gaan, Elucidating the thermal decomposition of dimethyl methylphosphonate by vacuum ultraviolet (VUV) Photoionization: pathways to the PO radical, a key species in flame-retardant mechanisms, Chem. - Eur. J. 21 (3) (2015) 1073–1080.
[8] S. Liang, P. Hemberger, J. Levalois-Grützmacher, H. Grützmacher, S. Gaan, Probing Phosphorus Nitride (P=N) and other elusive species formed upon pyrolysis of dimethyl phosphoramidate, Chem. - Eur. J. 23 (23) (2017) 5595–5601.
[9] Y. Ding, S.I. Stoliarov, R.H. Kraemer, Pyrolysis model development for a polymeric material containing multiple flame retardants: relationship between heat release rate and material composition, Combust. Flame 202 (2019) 43–57.
[10] Z. Xu, W. Xing, Y. Hou, B. Zou, L. Han, W. Hu, Y. Hu, The combustion and pyrolysis process of flame-retardant polystyrene/cobalt-based metal organic frameworks (MOF) nanocomposite, Combust. Flame 226 (2021) 108–116.
[11] S.I. Stoliarov, F. Raffan-Montoya, R.N. Walters, R.E. Lyon, Measurement of the global kinetics of combustion for gaseous pyrolyzates of polymeric solids containing flame retardants, Combust. Flame 173 (2016) 65–76.
[12] C. Ma, J. Wang, Y. Yuan, X. Mu, Y. Pan, L. Song, Y. Hu, An insight into gas phase flame retardant mechanisms of AHP versus AlPi in PBT: online pyrolysis vacuum ultraviolet photoionization time-of-flight mass spectrometry, Combust. Flame 209 (2019) 467–477.
[13] J. Miao, Y. Fang, X. Yang, Y. Zhu, A. Hu, G. Wang, Fabrication, flame retardancy and physical properties of phosphorus containing porous organic polymers/epoxy resin composites, Polym. Degrad. Stab. 176 (2020) 109159.
[14] https://www.flameretardants-online.com/flame-retardants/market.
[15] S. Yang, J. Wang, S. Huo, M. Wang, J. Wang, Preparation and flame retardancy of a compounded epoxy resin system composed of phosphorus/nitrogen-containing active compounds, Polym. Degrad. Stab. 121 (2015) 398–406.
[16] Y. Yuan, Y. Shi, B. Yu, J. Zhan, Y. Zhang, L. Song, C. Ma, Y. Hu, Facile synthesis of aluminum branched oligo(phenylphosphonate) submicro-particles with enhanced flame retardance and smoke toxicity suppression for epoxy resin composites, J. Hazard. Mater. 381 (2020) 121233.
[17] A.G. Shmakov, V.M. Shvartsberg, O.P. Korobeinichev, M.W. Beach, T.I. Hu, T. A. Morgan, Structure of a freely propagating rich CH4/air flame containing triphenylphosphine oxide and hexabromocyclododecane, Combust. Flame 149 (4) (2007) 384–391.
[18] L.H. Pham, H.D. Nguyen, J. Kim, D. Hoang, Thermal properties and fire retardancy of polypropylene/wood flour composites containing eco-friendly flame retardants, Fibers Polym. 20 (11) (2019) 2383–2389.
[19] O.P. Korobeinichev, S.A. Trubachev, A.K. Joshi, A. Kumar, A.A. Paletsky, A. G. Tereshchenko, A.G. Shmakov, R.K. Glaznev, V. Raghavan, A.M. Mebel, Experimental and numerical studies of downward flame spread over PMMA with and without addition of tri phenyl phosphate, Proc. Combust. Inst. 38 (3) (2021) 4867–4875.
[20] Y. Tan, Z.B. Shao, X.F. Chen, J.W. Long, L. Chen, Y.Z. Wang, Novel multifunctional organic–inorganic hybrid curing agent with high flame-retardant efficiency for epoxy resin, ACS Appl. Mater. Interfaces 7 (32) (2015) 17919–17928.
[21] R. Jian, P. Wang, W. Duan, J. Wang, X. Zheng, J. Weng, Synthesis of a Novel P/N/S-containing flame retardant and its application in epoxy resin: thermal property, flame retardance, and pyrolysis behavior, Ind. Eng. Chem. Res. 55 (44) (2016) 11520–11527.
[22] T. Yang, Y. Gao, X. Liu, X. Wang, B. Ma, Y. He, Flame-retardant polyamide 56 with high fire safety and good thermal performance, Polym. Adv. Technol. 33 (9) (2022) 2807–2819.
[23] L.A. Curtiss, K. Raghavachari, P.C. Redfern, A.G. Baboul, J.A. Pople, Gaussian-3 theory using coupled cluster energies, Chem. Phys. Lett. 314 (1) (1999) 101–107.
[24] C. Ao, S. Ruan, W. He, Y. Liu, C. He, K. Xu, L. Zhang, Toward high-level theoretical studies on the reaction kinetics of PAHs growth based on HACA pathway: an ONIOM[G3(MP2,CC)//B3LYP:DFT] method developed, Fuel 301 (2021) 121052.
[25] A.G. Shmakov, O.P. Korobeinichev, A.M. Mebel, D.P. Porfiriev, A.R. Ghildina, K. N. Osipova, D.A. Knyazkov, I.E. Gerasimov, Z. Liu, B. Yang, High-temperature thermal decomposition of triphenyl phosphate vapor in an inert medium: flow reactor pyrolysis, quantum chemical calculations, and kinetic modeling, Combust. Flame 249 (2023) 112614.
[26] Q. Hou, W. Li, J. Huang, C. Wang, J. Yang, Z. Serinyel, G. Dayma, W.J. Pitz, L. Zhao, F. Zhang, A combined theoretical and experimental study of the pyrolysis of pyrrolidine, Combust. Flame 258 (2023) 113063.
[27] Gaussian 16, Revision A.03, M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, G.A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A.V. Marenich, J. Bloino, B.G. Janesko, R. Gomperts, B. Mennucci, H.P. Hratchian, J.V. Ortiz, A.F. Izmaylov, J.L. Sonnenberg, D. Williams-Young, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, V.G. Zakrzewski, J. Gao, N. Rega, G. Zheng, W. Liang, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, K. Throssell, J.A. Montgomery, Jr., J.E. Peralta, F. Ogliaro, M.J. Bearpark, J.J. Heyd, E.N. Brothers, K.N. Kudin, V.N. Staroverov, T.A. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A.P. Rendell, J.C. Burant, S.S. Iyengar, J. Tomasi, M. Cossi, J.M. Millam, M. Klene, C. Adamo, R. Cammi, J.W. Ochterski, R.L. Martin, K. Morokuma, O. Farkas, J.B. Foresman, and D.J. Fox, Gaussian, Inc., Wallingford CT, 2016.